

Cross-cell-type transcriptomic transfer of combinatorial chemistry programs in Z-Screen

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Abstract

A scalable answer to “how will this compound behave in a cell type that was not directly assayed?” would change the economics of preclinical drug discovery. The current answer requires running every program in every cell type of interest, which scales poorly. Cell-intrinsic pharmacology is rarely identical across tissues, so a single reference cell line is often insufficient; running ten cell lines for every campaign is what teams do when they have to.

The canonical Z-Screen combinatorial-library transcriptomic dataset supports a useful first version of cross-cell-type projection, but only for comparisons with enough paired chemistry to evaluate honestly. We first define that eligible set, then benchmark held-out transfer at the level of individual molecules and grouped building-block programs in a 32-dimensional scVI latent space.

At the molecule level, direct reuse of the source-cell signature was inadequate, while learned transfer maps recovered target-cell information. In ZEL024 (HEK293 $\leftrightarrow$ H1650), centered-response cosine for HEK293 $\rightarrow$ H1650 improved from 0.146 with direct reuse to 0.279 with ridge transfer; the reverse direction reached 0.371. In ZEL031 (A549 $\leftrightarrow$ THP1), A549 $\rightarrow$ THP1 improved from 0.094 to 0.242. These molecule-level gains were real but modest, which motivated a chemistry-program view. Aggregating by chemistry-resolved building-block programs produced stronger transfer: in ZEL024, bb0+bb1 pair programs reached support-weighted cosine 0.863 for H1650 $\rightarrow$ HEK293 and 0.808 for HEK293 $\rightarrow$ H1650; in the two-position ZEL031 library, bb0 programs reached 0.714 for A549 $\rightarrow$ THP1 and 0.664 for THP1 $\rightarrow$ A549.

The strongest current claim is therefore not universal single-molecule prediction. It is that paired landmark data can translate chemistry-linked response programs across cell types when the program is defined at an aggregation level that still preserves chemical specificity. We deliberately demote attractive single-building-block results from ZEL028-2 because each centroid averages hundreds of distinct tuples, making the signal too broad to treat as a chemistry-resolved program. Cross-library projection was suggestive in a few directions but asymmetric and underpowered, and gene-level decoding should be read as hypothesis generation. The practical implication is operational: profile a new compound or chemical series once in an accessible source cell type and use a learned transfer map to prioritize target-cell follow-up, with the strongest current confidence at the level of grouped chemical programs rather than isolated molecules.

Significance

Every added cell type increases screening cost, and skipping the wrong cell type is how programs miss toxicity, selectivity, or therapeutically relevant biology. This paper presents a first practical route through that tradeoff. When the same chemistry is observed across paired cellular contexts, Z-Screen can learn transfer maps that recover target-cell signal better than direct reuse of the source profile. The strongest current unit is the grouped chemical program rather than the single molecule, which matches how early-stage discovery decisions are usually made: at the level of chemical series, not isolated compounds. For a discovery organization, this changes the budget calculus. Instead of running every program through every cell type of clinical interest, a smaller paired landmark atlas plus a single source measurement begins to do useful work for the rest.

Introduction

A substantial share of preclinical cost goes to running the same compound, panel, or library in every cell type of practical interest. The economics of early-stage discovery would change if a sufficiently broad paired landmark atlas allowed teams to:

- profile a new compound in one accessible source cell type,
- project its likely transcriptional response in unmeasured target cell types,
- and use those projections to prioritize efficacy work, counter-screen liabilities, and decide where wet-lab follow-up is worth the cost.

The right form of that claim is narrow but still important. Cross-cell-type transcriptomic transfer is a model for cell-intrinsic transcriptional response, and it does not substitute for pharmacokinetics, metabolism, tissue architecture, or organism-level toxicology. It is a practical decision-support primitive for the upstream half of discovery, where even partial accuracy is valuable.

The most directly comparable public reference points are L1000 [2], which provides a chemistry-side compound atlas in a small number of cell lines (mostly MCF7, PC3, A549, A375), and Tahoe-100M [3], which scales perturbational scRNA-seq across 50 cancer cell lines for approximately a thousand compounds. Both are valuable for cross-cell signature comparisons of the same compound, and neither provides combinatorial-chemistry provenance: a hit in L1000 or Tahoe-100M is a single profiled compound with no internal structure to its identity. scGeneScope [4] adds matched imaging but uses one cell line. Z-Screen, by contrast, uses one-bead-one-compound combinatorial libraries profiled in 50,000-well microchips by paired imaging and mRNA sequencing at 5 to 10 cells per well. That design is well suited to the cross-cell question because every well in the canonical dataset carries full chemistry provenance (building blocks plus full SMILES), the same combinatorial libraries are run across multiple cell lines, and the readout is a low-pass transcriptomic state for every compound rather than a sparse hit or non-hit label.

Compound-level overlap with these public atlases is also relevant for cross-cell positioning. Of 47 named Z-Screen control compounds, 14 (29.8%) are present in LINCS L1000 (palbociclib, sorafenib, crizotinib, ceritinib, rucaparib, veliparib, lapatinib, cobimetinib, GSK126, CP-673451, and others), with 3 of 4 Z-Screen cell lines (HEK293, A549, THP1) covered by L1000 sample

metadata. Tahoe-100M’s drug metadata covers 4 of the 47 named compounds (palbociclib, crizotinib, cobimetinib, capmatinib), with A549 as the only matched cell line. A direct cross-platform Spearman concordance test against LINCS L1000 level-5 consensus signatures (NCBI GEO GSE70138 and GSE92742), restricted to the 12,328-gene LINCS-visible universe, recovers positive correlation for all 11 (compound, cell_line) pairs with adequate coverage on both platforms. Median $\rho = 0.490$, mean = 0.393, with 8 of 11 above $\rho = 0.2$ and 5 reaching empirical $p = 0.048$ against an unmatched-compound permutation null (crizotinib, ZM-336372, sorafenib, rucaparib, veliparib). The HEK293 cell-line-pooled comparison separates matched from unmatched compound pairings at one-sided Mann-Whitney $p = 0.009$. Full overlap and concordance tables are in **Benchmarking/** in this repository.

With those ingredients, the cross-cell question can be posed in a controlled way: how much of a source-cell transcriptomic response can be projected into a target cell type using only a learned transfer map fit on a paired landmark subset. This paper characterizes how well that question is answered today, on data from a small number of pilot experiments, and defines what additional data would close the remaining gaps.

Results

Dataset coverage defines the eligible comparisons

The first question was not which model performs best. It was which cell-pair comparisons have enough overlapping chemistry to support a transfer claim at all. The canonical dataset is a pilot corpus with uneven depth across libraries and cell lines. ZEL024 has approximately 10,000 to 14,000 unique SMILES per cell line, including the densest system in the workspace: HEK293 at a median of 11 wells per compound. ZEL031 has 8,000 to 9,000 unique SMILES in A549 and THP1 at a median of 2 wells per compound. ZEL028-2 is broad but shallow, with 40,000 to 61,000 unique SMILES in A549 and HEK293 at a median of 1 well per compound.

That difference matters because a cross-cell benchmark needs the same chemistry in both source and target cells. At a minimum support filter of two cells, ZEL024 H1650 $\leftrightarrow$ HEK293 contributed 6,299 shared SMILES and ZEL031 A549 $\leftrightarrow$ THP1 contributed 3,561 shared SMILES. By contrast, ZEL028-2 cell-pair overlaps fell to 9 to 19 shared SMILES at the same filter, and ZEL031 comparisons involving H1650 were similarly too small. At a five-cell filter, the eligible set narrowed further: ZEL024 H1650 $\leftrightarrow$ HEK293 retained 353 shared SMILES, and ZEL031 A549 $\leftrightarrow$ THP1 retained 140.

These coverage facts set the order of the evidence. Molecule-level transfer can be tested most cleanly in ZEL024 and ZEL031. ZEL028-2 can still contribute to grouped and cross-library exploratory analyses, but its singleton-heavy structure prevents it from carrying the main per-molecule claim.

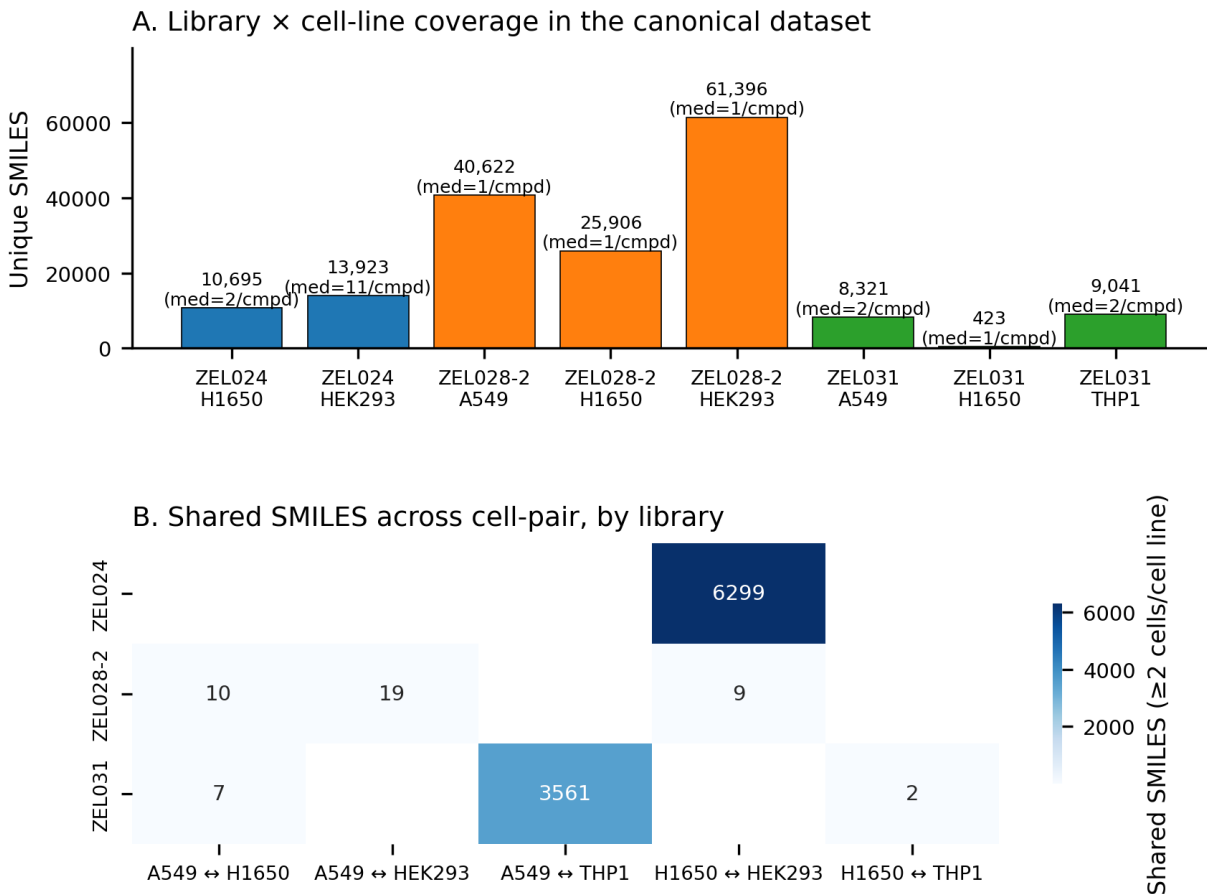


Figure 1: Figure 1. Library-by-cell-line coverage in the canonical Z-Screen dataset. Top: unique SMILES and median wells per compound per system. Bottom: shared-SMILES counts across cell-pair directions, by library. The two dense paired systems used in the molecule-level transfer benchmark are ZEL024 (H1650 $\leftrightarrow$ HEK293, 6,299 shared) and ZEL031 (A549 $\leftrightarrow$ THP1, 3,561 shared).

Molecule-level transfer is real in dense paired libraries

The molecule-level benchmark used held-out individual molecules in the two cell-pair systems with sufficient shared SMILES coverage:

- ZEL024, H1650 $\leftrightarrow$ HEK293, 6,299 shared molecules,
- ZEL031, A549 $\leftrightarrow$ THP1, 3,561 shared molecules.

Each molecule was represented by the mean of its 32-dimensional scVI latent coordinates within a given library and cell line. Direct reuse of the source-cell signature (identity) was compared against learned source-to-target transfer models. The centered-response benchmark is the primary readout because matched vehicle controls are unavailable in the canonical dataset, and the centered formulation is the closest available proxy for perturbational response. Under that benchmark, learned transfer consistently outperformed direct reuse at the largest training sets:

- ZEL024, HEK293 -> H1650, train 5,000: identity 0.146, ridge 0.279, PLS 0.275.
- ZEL024, H1650 -> HEK293, train 5,000: identity 0.146, ridge 0.371, PLS 0.358.
- ZEL031, A549 -> THP1, train 3,000: identity 0.094, ridge 0.242, PLS 0.242.
- ZEL031, THP1 -> A549, train 3,000: identity 0.094, ridge 0.212, PLS 0.214.

Two readings follow. Direct source-signature reuse alone is inadequate, so cross-cell projection is a real prediction problem rather than an artifact of cell-state offset. A simple regularized global linear or low-rank map already captures useful molecule-specific information once enough paired examples are available. The gains are also modest in absolute terms, especially in centered response space. This is the key reason to ask whether chemistry-resolved groups, rather than isolated molecules, are the more stable unit of transfer in the current dataset.

The raw-profile benchmark is reported in the supplementary material as a diagnostic. Performance is much higher in raw space (for example, ZEL024 HEK293 -> H1650 reaches 0.546 with ridge transfer at train 5,000, versus 0.063 with direct reuse), and a substantial portion of that gain reflects baseline cell-state translation rather than perturbational response. The centered-response benchmark above is the cleaner main readout for a discovery use case, because it removes the constant offset that exists between two cell lines regardless of any particular perturbation.

Molecule-level cross-cell transfer in centered response space

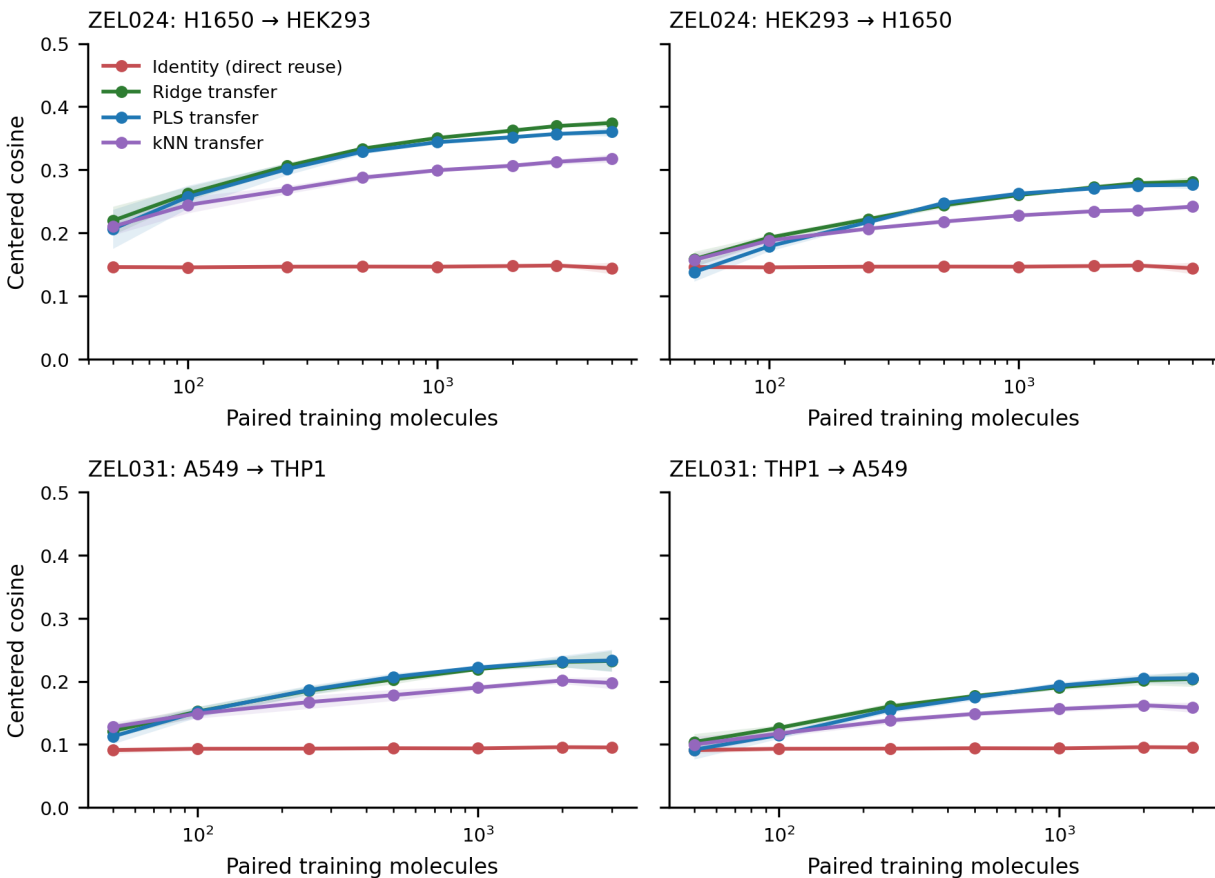


Figure 2: Figure 2. Molecule-level cross-cell transfer in centered response space across four well-supported cell-pair directions in ZEL024 and ZEL031. Direct reuse of the source signature is consistently weaker than learned transfer; ridge and PLS transfer are nearly tied. The absolute centered-response cosines remain modest, motivating the grouped-program analysis that follows.

Chemistry-resolved grouping exposes stronger transferable programs

Single-molecule signatures are noisy because each compound is often represented by a small number of wells and by a low-pass transcriptomic readout. The next test therefore asked whether source-cell and target-cell responses align more cleanly after averaging molecules that share a defined combinatorial chemistry program.

Reporting is restricted to aggregation levels that remain chemically interpretable. For ZEL024, which has four substantive building-block positions, the reported program is a BB-pair (bb0+bb1); each pair-level signature averages roughly 129 to 166 distinct tuples rather than collapsing the whole library. For ZEL031, which has only two substantive positions, a single bb0 group is the closest analog because the other position remains the varying partner chemistry.

Under those chemistry-resolved aggregations, grouped transfer was substantially stronger than the

within-cell direct-reuse baseline:

- ZEL024, H1650 $\rightarrow$ HEK293, grouped by bb0+bb1: 84 shared pairs, median support 240.5 wells, identity 0.471, ridge transfer **0.863**.
- ZEL024, HEK293 $\rightarrow$ H1650, grouped by bb0+bb1: 84 shared pairs, median support 240.5 wells, identity 0.471, ridge transfer **0.808**.
- ZEL031, A549 $\rightarrow$ THP1, grouped by bb0 in a two-position library: 129 shared groups, median support 157 wells, identity 0.419, ridge transfer **0.714**.
- ZEL031, THP1 $\rightarrow$ A549, grouped by bb0 in a two-position library: 129 shared groups, median support 157 wells, identity 0.419, ridge transfer **0.664**.

This is the strongest current result because it answers the limitation exposed by the molecule-level benchmark. The claim is not that arbitrary unseen molecules are solved. It is that chemistry-resolved building-block programs define reusable response units that transfer across cell types more cleanly than sparse individual molecules do. That is a practical unit for early discovery, where teams often advance or deprioritize chemical series before they have enough evidence to treat every analog as an independent biological object.

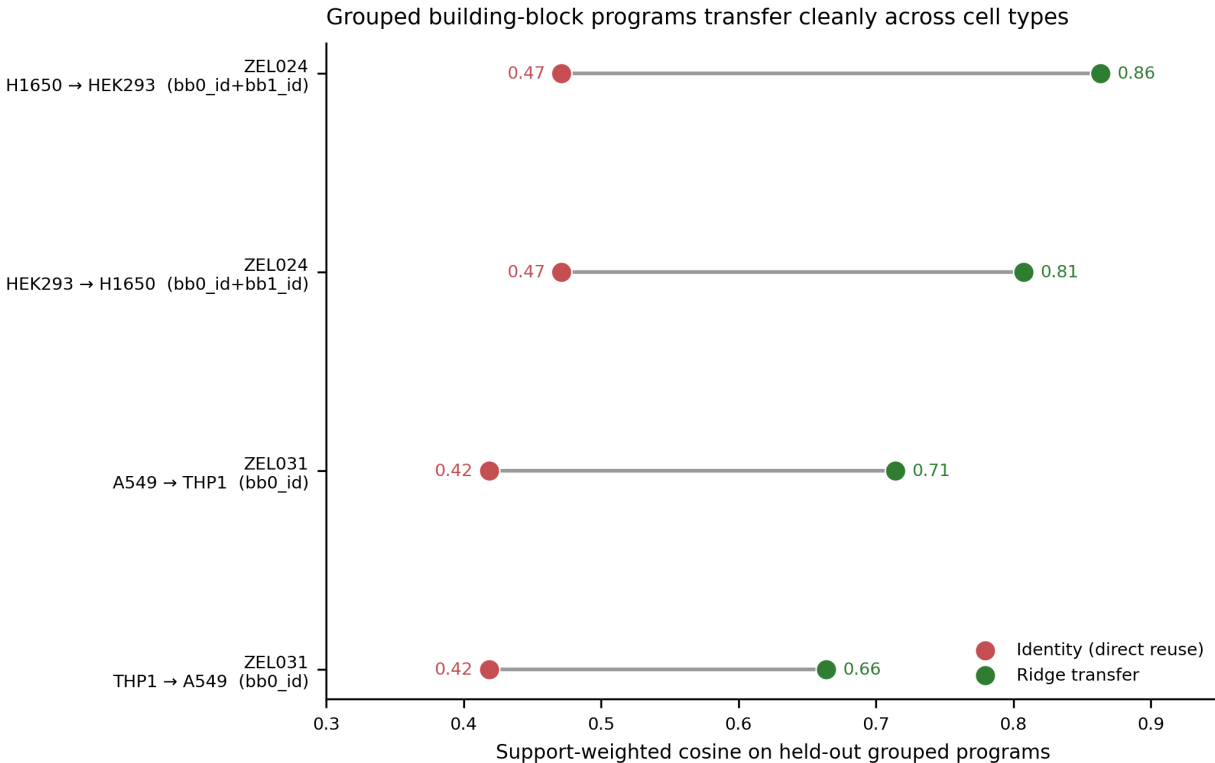


Figure 3: Figure 3. Grouped building-block program transfer at chemistry-resolved aggregation levels (BB-pair for ZEL024, single position bb0 for the two-position ZEL031 library). Identity (direct reuse of the source-cell signature) sits in the 0.42 to 0.52 range, while a regularized learned transfer map reaches support-weighted cosine 0.66 to 0.86 across four chemistry-resolved program-and-direction combinations.

The same point is visible when the molecule-level and grouped-program results are placed side by side. At the molecule level, learned transfer roughly doubles the centered cosine over identity in ZEL031 (0.094 to 0.21 to 0.24) and lifts it by 1.9 to 2.5 times in ZEL024. At the chemistry-resolved grouped-program level (ZEL024 BB-pair, ZEL031 bb0), learned transfer increases support-weighted cosine by 0.25 to 0.39 absolute over identity, with all four chemistry-resolved directions ending in the 0.66 to 0.86 cosine range. The grouped-program unit is the stronger current asset and is the unit recommended for early use in discovery decisions.

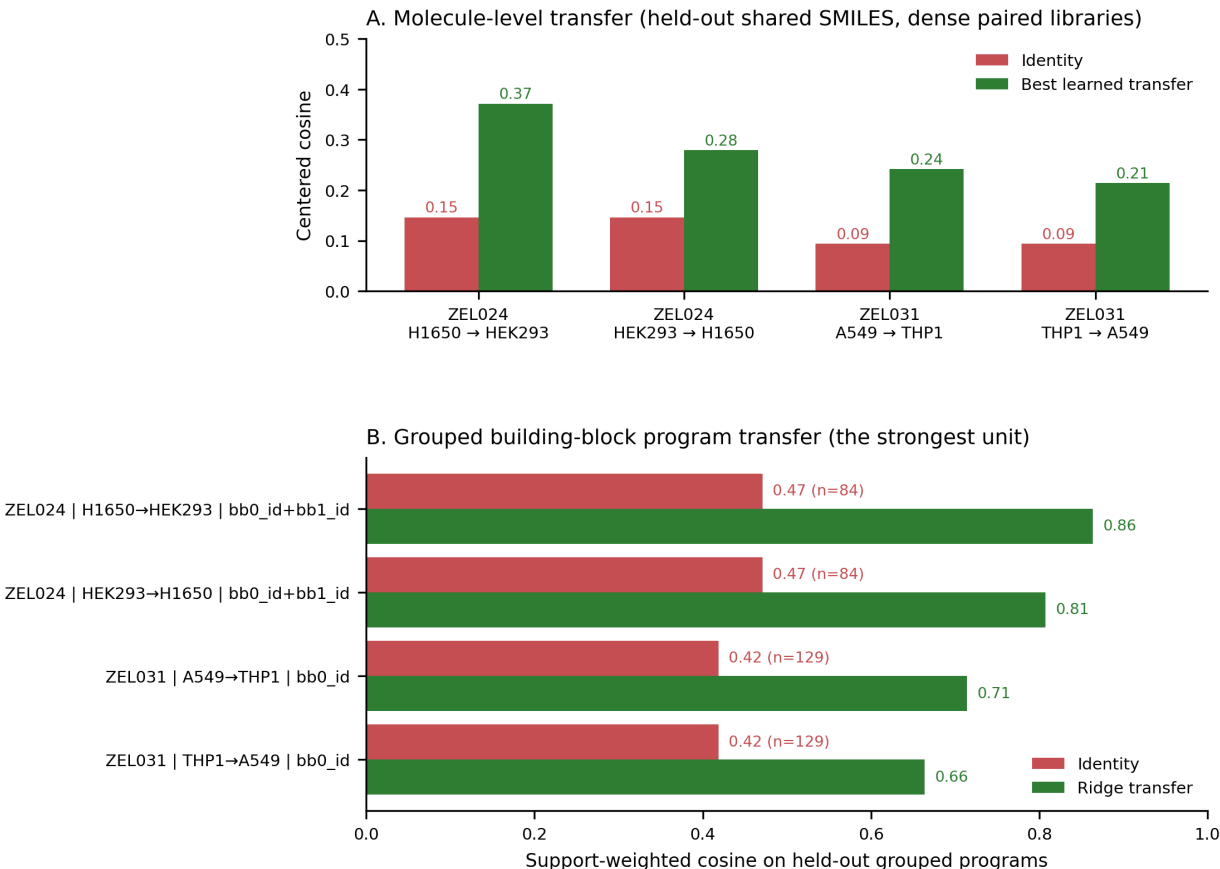


Figure 4: Figure 4. Best learned transfer versus identity at both molecule level (top) and grouped building-block program level (bottom). Learned transfer wins at every cell-pair direction tested, with the largest gains at the grouped-program level.

Guardrail: over-aggregated single-BB results should not headline

Some apparently attractive grouped results fail the chemical-specificity test. In ZEL028-2, single-building-block transfer reached high weighted cosines in the full grouped-analysis table: for example, bb1 grouping reached 0.837 for H1650 → HEK293 and 0.850 for A549 → HEK293. Those numbers are not reported as main results because ZEL028-2 has four substantive building-block positions. A single-position centroid therefore averages over broad chemistry, while the more specific BB-pair rows are much shallower per group.

That aggregation is useful as a guardrail, not as a headline. It says that broad library-level and cell-pair structure can be transferred, but it no longer isolates a chemistry-resolved program in the way a ZEL024 BB-pair or ZEL031 bb0 group does. Reporting those ZEL028-2 values beside the chemistry-resolved results would make the model look stronger while making the biological claim less precise.

Table 1. ZEL028-2 grouped-transfer guardrail. Median support is the minimum matched-cell support across source and target for each grouped centroid.

Direction	Group	Class	Groups	Median support	Transfer	Gain	Interpretation
A549 -> HEK293	bb1_id	single-BB aggregate	86	526.5	0.850	0.332	high transfer but over-aggregated
H1650 -> HEK293	bb1_id	single-BB aggregate	86	333.5	0.837	0.366	high transfer but over-aggregated
HEK293 -> A549	bb1_id	single-BB aggregate	86	526.5	0.762	0.244	high transfer but over-aggregated
A549 -> HEK293	bb1_id+bb1_id pair group	BB pair group	6,690	5.0	0.227	0.100	more specific but shallow per group
A549 -> HEK293	bb3_id+bb3_id pair group	BB pair group	7,158	5.0	0.215	0.089	more specific but shallow per group
HEK293 -> A549	bb1_id+bb1_id pair group	BB pair group	6,690	5.0	0.213	0.086	more specific but shallow per group

Cross-library projection is suggestive, asymmetric, and underpowered

A further question is whether a centered source-to-target transfer matrix learned in one library can project molecules from a different library measured in the same cell-pair direction. The answer is mixed.

Several projections were directionally positive:

- ZEL024 -> ZEL028-2, HEK293 -> H1650: weighted cosine 0.033 -> 0.059.
- ZEL024 -> ZEL028-2, H1650 -> HEK293: 0.033 -> 0.055.
- ZEL028-2 -> ZEL031, A549 -> H1650: 0.120 -> 0.216.

Other directions were weak or negative:

- ZEL028-2 -> ZEL024, HEK293 -> H1650: 0.119 -> 0.067.
- ZEL031 -> ZEL028-2, A549 -> H1650: 0.103 -> -0.009.

The pattern is not strong enough to support a clean dense-to-sparse rule. ZEL024 trained maps did improve transfer into ZEL028-2 in both HEK293/H1650 directions, but ZEL028-2 also improved one projection into ZEL031 while failing in the reverse ZEL024 direction. The safer interpretation is that a source-to-target cell transformation can sometimes survive a library change, but the present cross-library tests are dominated by small overlaps, singleton-heavy target sets, and disjoint building-block vocabularies. Cross-library projection is therefore the next prospective dataset to design, not a settled generalization result.

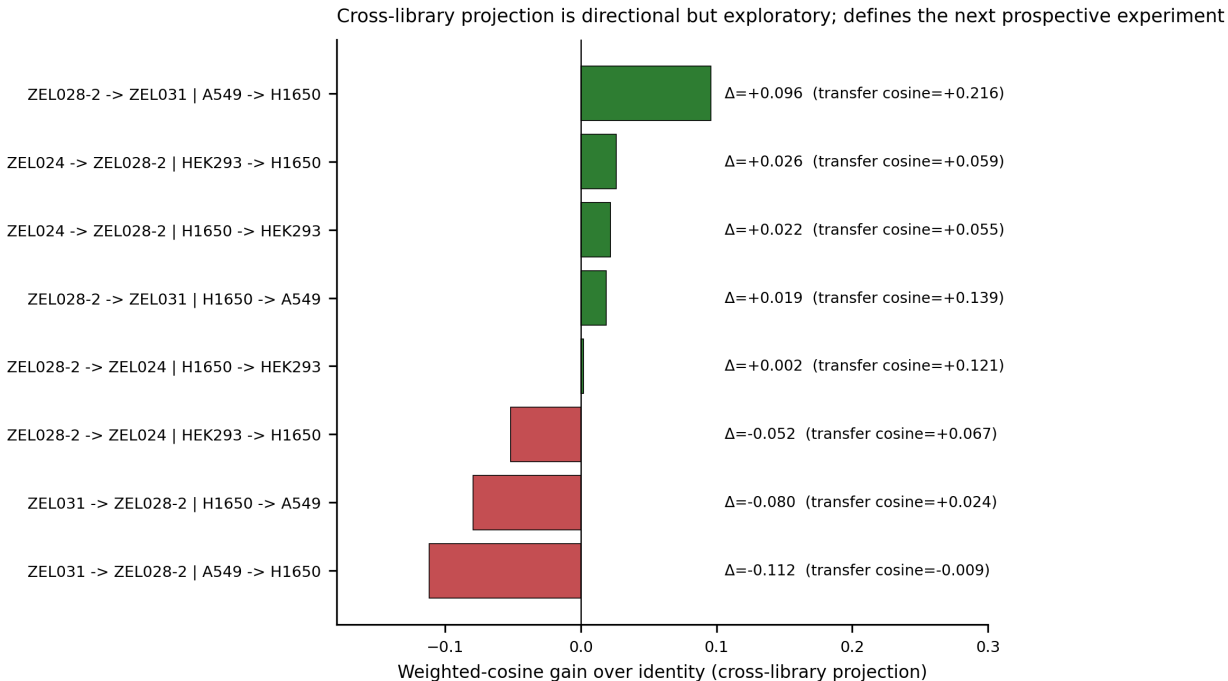


Figure 5: Figure 5. Cross-library projection results. Some cross-library projections are positive, but performance remains small, asymmetric, and sensitive to which library acts as teacher. The result is best read as a design brief for a prospective cross-library dataset.

Exploratory gene-level decoding partially recovers held-out target signatures

A separate experiment tested whether the learned cell-transfer map can be lifted back from the 32-dimensional latent space into ranked gene-level hypotheses. This was treated as an exploratory decoder experiment rather than a main claim. For each dense paired direction, a source-to-target latent transfer map was fit on shared training compounds, a target-cell ridge decoder was fit from target latents to 800 training-selected variable genes, and held-out target-cell gene deltas were compared against measured pseudobulk deltas. The strongest direction was ZEL024 H1650 $\rightarrow$ HEK293: transferred latents reached mean per-compound gene-rank Spearman 0.182 and top-50 absolute-delta overlap 0.139, compared with 0.021 and 0.083 when the source latent was passed directly through the target decoder. The reverse ZEL024 direction and both ZEL031 directions were weaker, with transfer Spearman values of 0.073 to 0.098.

The result is useful as hypothesis generation. A transferred latent can nominate genes or pathways to inspect in the target cell type, but the decoder still depends on observed target-cell training profiles and the correlations are not high enough to claim full transcriptome reconstruction.

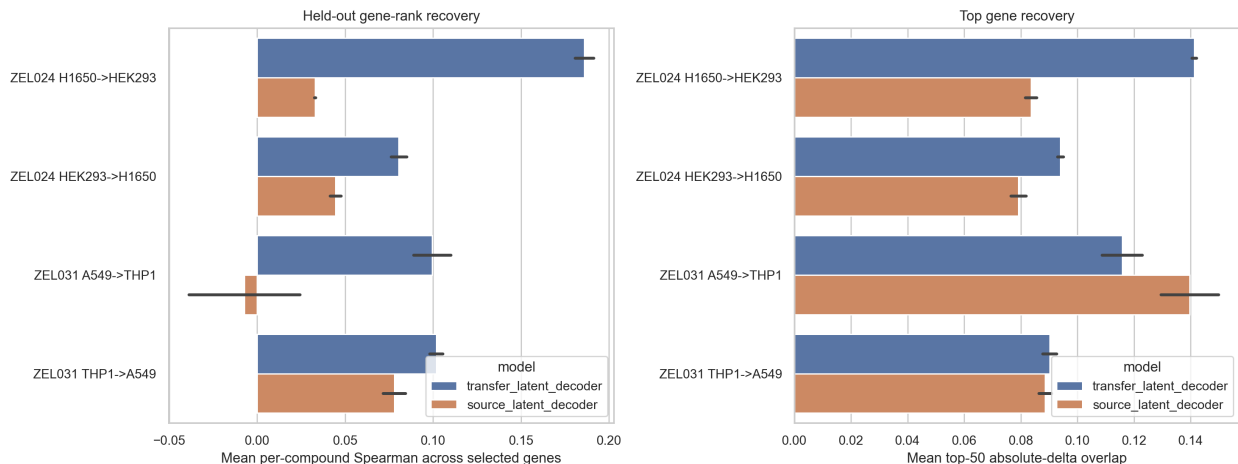


Figure 6: Supplementary Figure 2. Exploratory gene-level transfer. A target-cell decoder composed with the learned source-to-target latent map partially recovers held-out gene-rank structure, with the clearest gain in ZEL024 H1650 $\rightarrow$ HEK293 and weaker gains in the remaining directions.

Discussion

The current data support a two-level view of cross-cell transfer. At the molecule level, paired transfer is real in the eligible dense libraries: even after centering within cell line, learned transfer maps recover target-cell information that direct source reuse misses. The absolute molecule-level signal is still modest. At the chemistry-resolved grouped-program level (BB-pair in dense ZEL024; single-position bb0 in the two-position ZEL031 library), transfer becomes substantially stronger, with weighted cosine in the 0.66 to 0.86 range across four cell-pair directions. From a discovery perspective, the encouraging reading is that the first economically useful layer of cross-cell prediction is chemistry-program transfer rather than universal single-molecule prediction.

The current results also argue against an overly strong “one universal matrix solves everything” framing. ZEL028-2 shows why aggregation discipline matters: broad single-building-block centroids can produce appealing values while no longer representing a chemistry-resolved program. Cross-library projection remains asymmetric and fragile in sparse settings, and a more durable long-term architecture is likely a shared perturbational latent space with cell-type-specific decoders or adapters, coupled to an uncertainty model that rejects out-of-domain projections. That architecture fits naturally on top of the present transfer maps without replacing them, and it is consistent with the direction the public field is taking on cross-cell perturbation modeling.

In practical terms, a paired landmark atlas plus one source-cell measurement is enough to start prioritizing target-cell transcriptomic follow-up for the same chemistry, with the strongest current confidence at the level of grouped chemical programs rather than individual molecules. The immediate use cases are counter-screen prioritization, cross-cell-line selectivity ranking, and decision support for which downstream cell types deserve direct profiling. The next dataset should be designed around prospective cross-library scaffold-family transfer with enough repeated chemistry to distinguish true transfer from sparse-overlap variance.

Methods

Data inputs

All analyses use the canonical Z-Screen RNA aggregate dataset, specifically:

- `canonical_obs.parquet`,
- `canonical_scvi_latents.parquet`,
- `canonical_chemistry.parquet` for grouped building-block benchmarks.

Only rows with a non-null `smiles_hash` (the public-bundle stand-in for a chemistry-resolved compound identity) and `chemistry_grain == "building_block_annotation"` were retained. Compound-control-only rows were not used for the transfer benchmark because their annotations are mechanism-of-action labels rather than chemistry coordinates.

scVI latent space rather than full gene expression

The transfer benchmarks are built on 32-dimensional scVI latent coordinates rather than full gene-level vectors, for two reasons. First, low-pass per-well RNA depth in the present dataset (each well averages 5 to 10 cells, sequenced as one pseudobulk row) means individual genes are dominated by sampling noise; scVI absorbs that noise into shared structure across observations. Second, the latent is comparable across cell types to a degree that raw expression is not, which is the property the cross-cell transfer experiments require. The exploratory gene-level decoder experiment is included as a separate analysis specifically to assess how much of the latent gain survives when the result is lifted back to gene level.

Centered-response as the primary metric

Two cell lines have different baseline transcriptional states regardless of any perturbation. A cross-cell predictor that “knows” the constant baseline shift will show high raw-space cosine even if it has learned nothing about perturbational response. Centering removes that confound. For each train/test split, source and target training means were subtracted before fitting and evaluation. This is the closest available proxy for a vehicle-control normalization in a dataset that does not have matched vehicle controls.

Molecule-level transfer benchmark

For each (library, cell line, SMILES), scVI latent coordinates were averaged across wells. Shared molecules between a source and target cell line were then benchmarked under two evaluation spaces, `raw_profile` and `centered_response`. The models compared were `identity`, `global_shift` (raw space only), `ridge_transfer`, `pls_transfer`, and `knn_transfer`. The `global_shift` baseline was included so that the bulk-offset component of cross-cell variance is visible separately from chemistry-specific signal. Performance was measured by row-wise cosine similarity on held-out molecules and aggregated as mean cosine across multiple train-test splits.

Grouped building-block transfer

Within each library and cell-pair direction, molecules were grouped by candidate building-block keys (`bb0_id`, `bb1_id`, or multi-column combinations such as `bb0_id+bb1_id`). For each shared grouped key, source and target centered latent signatures were averaged and benchmarked using a centered ridge transfer model across repeated train-test splits. Performance was summarized as support-weighted cosine similarity across grouped programs. Support weighting prevents a few groups with thousands of supporting wells from being drowned out by many groups with five.

Cross-library transfer

For a fixed cell-pair direction, a centered ridge transfer matrix was trained in one library and applied to molecules from another library. Because the sparser libraries are highly singleton-heavy, adaptive minimum-well thresholds were used so that denser libraries retained stricter filters while still allowing exploratory sparse projections. Directional asymmetry is reported directly rather than corrected away, because teacher-library depth and target-library support are part of the question.

Exploratory gene-level transfer decoder

For the four dense paired directions, held-out shared SMILES were evaluated with a composed latent-to-gene workflow. A centered ridge model first mapped source-cell scVI centroids into target-cell scVI space. Separately, a target-cell ridge decoder was trained on the training compounds to map target-cell scVI centroids to log-CPM pseudobulk deltas over 800 variable genes selected from target training cells. Held-out predictions were scored by per-compound Pearson and Spearman correlation over gene deltas and by top-50 / top-100 absolute-delta gene overlap.

Limitations

- Matched vehicle controls are unavailable in the canonical dataset, so the centered-response benchmark is the closest available proxy for perturbational response rather than a fully DMSO-normalized effect.
- The strongest molecule-level benchmark exists only for the denser paired libraries ZEL024 and ZEL031; ZEL028-2 becomes too sparse under a strict per-molecule support filter.
- Cross-library transfer is limited by sparse sampling and by disjoint building-block vocabularies across libraries, which prevents a clean cross-library tuple-transfer formulation today.
- All analyses are based on averaged 32-dimensional scVI latent signatures rather than full transcript-level models or time- and dose-aware transfer.
- The gene-level decoder experiment is exploratory and uses a target-cell decoder trained on observed target profiles; it should be read as a way to generate ranked gene hypotheses from transferred latents, not as evidence that full transcriptomes can already be prospectively reconstructed from an unmeasured target cell state.
- The current dataset reflects a small number of pilot experiments; a prospective cross-library transfer study should be designed with repeated chemistry and predefined eligibility thresholds.

Data availability

Data tables, manuscript figures, and analysis inputs needed to reproduce this manuscript are organized in this repository under `paper4a/` and `data/ZScreen_Canonical_Dataset/`. A persistent-archive deposition with an assigned DOI will accompany the corresponding preprint posting.

Code availability

Analysis and figure-generation scripts are in `paper4a/scripts/`. Package-level dependencies are in the repository root `requirements.txt`. Tested with Python 3.14.3 on Windows.

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